

MongoDB

DAY (1 to 2)	Topic	Status (Completed)	Rating (1-10)	Comment
1	Understanding the differences between SQL and NoSQL databases Overview of document-based databases MongoDB Overview 1)Features and advantages of MongoDB 2)BSON data format and its relation to JSON 3)MongoDB data structure: databases, collections, and documents			
	Operations in MongoDB			
	Insert			
	Update			
	Delete			
	Read			
2	Practical			

Practical

DAY (1 to 2)	Practical Questions	Status (Completed)
2	1. Set up a MongoDB database named "CRUD_DB".	
	2. Create a collection named "users" within the "CRUD_DB" database.	
	3. Insert at least 5 documents into the "users" collection, each representing a user with fields such as name, age, email, and country.	
	4. Ensure that each document has a unique identifier.	
	5. Retrieve and display all documents from the "users" collection.	
	6. Find and display users who are between 25 and 35 years old.	
	7. Find and display users who are from a specific country (e.g., "USA").	
	8. Update the email address of a user.	
	9. Increment the age of users who are above 30 by 1 year.	
	10. Delete a user document from the collection.	
	11. Delete all users who are below 25 years old.	
	12. Set up a MongoDB database named "Product_DB".	
	13. Create a collection named "products" within the "Product_DB" database.	
	14. Insert at least 5 documents into the "products" collection, each representing a product with fields such as name, category, price, and quantity.	
	15. Ensure that each product has a unique identifier.	
	16. Retrieve and display all documents from the "products" collection.	
	17. Find and display products within a specific price range (e.g., between \$20 and \$50).	
	18. Find and display products in a particular category (e.g., "Electronics").	
	19. Update the price of a product.	
	20. Decrement the quantity of products that are currently in stock.	
	21. Delete a product document from the collection.	
	22. Delete all products that are out of stock.	

Node

DAY (1 to 3)	Topic	Status (Completed)	Rating (1-10)	Comment
1	Introduction to Node JS • Introduction • What is Node JS? • Advantages of Node JS			
	Node JS Modules • Functions • Module, Module Types(Core Modules, Local Modules) • Module.Exports			
	Node Package Manager • What is NPM • Installing Packages Locally • Adding dependency in package.json • Installing packages globally			
2	Creating Web server • Creating web server • Handling http requests • Sending requests			
	- File System • Reading File • Writing a File • Opening a file • Deleting a file			
3	Practical			
4	Database Integration using Mongoose			

DAY (1 to 2)	Practical Questions	Status (Completed)
1	1. Create a module named mathOperations.js that exports functions for addition, subtraction, multiplication, and division. Import this module into a separate calculator.js file and demonstrate the usage of these operations by performing calculations (e.g., add 5 and 3, multiply 10 and 2).	
	2. Develop module for a) Read File b) Write File c) Delete File d) Make Directory e) Remove Directory.	
	3. Create a module named httpServer.js that exports a function to start an HTTP server using the built-in http module in Node.js. Import this module into a separate file and demonstrate starting the server on a specific port, handling incoming requests, and sending a response.	
	4. Create basic route based handling a) Create routes for index, about, contact b) Define corresponding logic what page will be displayed to it in the end.	
	5. Provide error handling for wrong routes such as a) 404 page not found	
2	6. Demonstrate Connectivity between node and mongodb using mongoose library perform following operation. a) Insert. b) Delete. c) Update. d) Find.	

EXPRESS JS

Sr no.	Topics	Status	Completed
1	Difference between framework and library		
	Introduction to Express-What/Why/How/Features		
	Introduction to http: what/why/how		
	Types of http methods -get/post/put/delete/push/patch		
	Restful API		
	creating server using express and built restful api		
	Practicals		
2	Introduction to middleware		
	Types of middleware: -Built in middleware -User defined middleware		
	Practicals		
3	Introduction to EJS		
	Tags in EJS		
	conditional statements and loops		
	Dynamic Content Rendering		
	Practicals		
4	Introduction to MVC Architecture		
	Creating Restful API using MVC architecture, mongoose db connectivity, and routing		
5	Practicals		

EXPRESS PRACTICAL QUESTIONS

Session no.	Practicals	Status
1	Create a web application using express	
	Create a RESTful API to display the Home and About Us pages.	
2	Create a web application and RESTful API using middleware such as <code>json()</code> , <code>urlencoded()</code> , and <code>static()</code> .	
	Create a RESTful API with HTTP methods (GET, POST, PATCH, DELETE) for an Employees module using Mongoose for database connectivity and Thunder Client to test the API. Use middleware such as <code>json()</code> , <code>urlencoded()</code> , and <code>static()</code> where appropriate.	
3	Create an MVC structure for a RESTful API with HTTP methods (GET, POST, PATCH, DELETE) for an Employees module using Mongoose for database connectivity.	
4	Write a "Hello World" program using EJS templating with Express.	
	Create a <code>show.ejs</code> file that displays data retrieved from a database using EJS templating.	
5	Create a Blog Application using MVC architecture with Express for server-side routing, Mongoose for database connectivity, and EJS for templating.	

REACT JS

SR NO.	Topic	Status (Completed)	Rating (1-10)	Comment
1	JS Fundamentals-Variables, Data Types, and Operators · Functions (Arrow functions, Function declarations, Function expressions) · Arrays and Array Methods (map, filter, reduce, etc.) · Objects and Object-Oriented Programming (OOP) concepts · ES6 Features (Destructuring, Spread and Rest operators, Classes, Promises, etc.)			
	SPA MPA			
	software components			
	Library framework difference			
	Introduction to react js			
	JSX			
	Component & Component Architecture			
2	FunctionalComponent			
	Class Components			
	Props			
2	Practicals			
3	useState Hooks			
	EventHandling			
	Conditional Rendering			
	Css			
	List Mapping			
3	Practicals			
4	Form Handling			
	Introduction to Hooks			
	useState Hooks			
	useEffect Hooks			
	Axios			
	Fetch Multiple and single data from json place holder server			
5	Practicals			
6	useContext			
	useReducer			
6	Practicals			
7	Routing			
	Mern Project (Frontend + Backend)			
	Practicals			